

CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Subsidiary Level and GCE Advanced Level

MARK SCHEME for the May/June 2014 series

9702 PHYSICS

9702/23

Paper 2 (AS Structured Questions), maximum raw mark 60

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International Examinations

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- 1 (a) current, mass and temperature
two correct 2/2, one omission or error 1/2 A2 [2]
- (b) σ : no units, V : m^3
 E_P : $\text{kg m}^2 \text{s}^{-2}$
 C : $\text{kg m}^2 \text{s}^{-2} \times \text{m}^{-3} = \text{kg m}^{-1} \text{s}^{-2}$ C1
C1
A1 [3]
- 2 (a) scalar has magnitude only B1
vector has magnitude and direction B1 [2]
- (b) (i) $v^2 = 0 + 2 \times 9.81 \times 25$ (or using $\frac{1}{2} m v^2 = mgh$) C1
 $v = 22(.1) \text{ m s}^{-1}$ A1 [2]
- (ii) $22.1 = 0 + 9.81 \times t$ (or $25 = \frac{1}{2} \times 9.81 \times t^2$) M1
 $t (=22.1/9.81) = 2.26 \text{ s}$ or $t [= (5.097)^{1/2}] = 2.26 \text{ s}$ A0 [1]
- (iii) horizontal distance $= 15 \times t$
 $= 15 \times 2.257 = 33.86$ (allow $15 \times 2.3 = 34.5$) C1
(displacement) $^2 = (\text{horizontal distance})^2 + (\text{vertical distance})^2$ C1
 $= (25)^2 + (33.86)^2$ C1
displacement $= 42$ (42.08) m (allow 43 (42.6) m, allow 2 or more s.f.) A1 [4]
- (iv) distance is the actual (curved) path followed by ball B1
displacement is the straight line/minimum distance P to Q B1 [2]
- 3 (a) work done is the product of force and the distance moved in the direction of the force
or product of force and displacement in the direction of the force B1 [1]

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- (b) (i) work done equals the decrease in GPE – gain in KE B1 [1]
- (ii) 1. distance = area under line
 $= (7.4 \times 2.5) / 2 = 9.3 \text{ m (9.25 m)}$ C1
M1 [2]
- or
- acceleration from graph $a = 7.4 / 2.5 (= 2.96)$ (C1)
and equation of motion $(7.4)^2 = 2 \times 2.96 \times s$ gives $s = 9.3 (9.25) \text{ m}$ (A1)
2. kinetic energy $= \frac{1}{2} m v^2$ C1
 $= \frac{1}{2} \times 75 \times (7.4)^2$ C1
 $= 2100 \text{ J}$ A1 [3]
3. potential energy $= mgh$ C1
 $h = 9.3 \sin 30^\circ$ C1
 $PE = 75 \times 9.81 \times 9.3 \sin 30^\circ = 3400 \text{ J}$ A1 [3]
4. work done = energy loss C1
 $R = (3421 - 2054) / 9.3$ C1
 $= 150 (147) \text{ N}$ A1 [3]
- 4 (a) add small mass to cause extension then remove mass to see if spring returns to original length M1
repeat for larger masses and note maximum mass for which, when load is removed, the spring does return to original length A1 [2]
- (b) Hooke's law requires force proportional to extension B1
graph shows a straight line, hence obeys Hooke's law M1 [2]
- (c) $k = \text{force} / \text{extension}$ C1
 $= (0.42 \times 9.81) / [(30 - 21.2) \times 10^{-2}]$ C1
 $= 47 (46.8) \text{ N m}^{-1}$ A1 [3]
- 5 (a) lost volts/energy used within the cell/internal resistance B1
when cell supplies a current B1 [2]

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- (b) (i) $E = I(R + r)$
 $4.5 = 0.65 (6.0 + r)$
 $r = 0.92 \Omega$ C1
A1 [2]
- (ii) $I = 0.65 \text{ (A)}$ and $V = IR$
 $V = 0.65 \times 6 = 3.9 \text{ V}$ C1
A1 [2]
- (iii) $P = V^2/R$ or $P = I^2R$ and $P = IV$
 $= (3.9)^2/6 = 2.5 \text{ W}$ C1
A1 [2]
- (iv) efficiency = power out/power in
 $= I^2R/I^2(R + r) = R/(R + r) = 6.0/(6.0 + 0.92) = 0.87$ C1
A1 [2]
- (c) (circuit) resistance decreases B1
current increases M1
more heating effect A1 [3]
- 6 (a) (i) progressive wave transfers energy, stationary wave no transfer of energy/
keeps energy within wave B1 [1]
- (ii) (progressive) wave/wave from loudspeaker reflects at end of tube B1
reflected wave overlaps (another) progressive wave B1
same frequency and speed hence stationary wave formed B1 [3]
- (iii) (side to side) along length of tube/along axis of tube B1 [1]
- (b) all three nodes clearly marked with N/clearly labelled at cross-over points B1 [1]
- (c) phase difference = 0 A1 [1]
- (d) (i) $v = f\lambda$ C1
 $\lambda = 330/440 = 0.75 \text{ m}$ A1 [2]
- (ii) $L = 5/4 \lambda$ C1
 $= 5/4 \times 0.75 = 0.94 \text{ m}$ A1 [2]